Can Agentic AI Settle a 20-Year-Old Open Problem?

The decidability of $\mathcal{ALCI}_{\mathrm{RCC5}}$ — a machine-checked campaign

Michael Wessel

with AI assistants: Claude Opus 4.6–4.8, Fable 5, Opus 5 (Anthropic);
GPT-5.4, 5.4 Pro, 5.5, 5.6 Pro, 5.6 Sol (OpenAI)

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The one-slide version

- In 2002/2003 I defined a family of spatial description logics and left their central decidability question open. It still is.
- In 2026 I directed a five-month AI campaign on it: frontier models, adversarial review, machine verification.
- The full question is not settled; we know exactly where our main route is stuck (a width property, F6).
- A sub-question is settled: drop ∀PO and the logic is decidable, by a procedure the Lean kernel checks.

How to read everything in this talk
The mathematics is AI-produced and *unverified by human experts*. Standing label for the full logic: **strongly supported, *not* certified** — but the ∀PO-free fragment is **machine-certified**, reviewed three times.

Five ways two regions can relate (RCC5)



- Every pair of regions stands in exactly one of these five relations.
- *Jointly exhaustive, pairwise disjoint* — the classic qualitative spatial calculus (a coarser cousin of RCC8).
- Think GIS, layout, biology, any "what contains / touches what" domain — no coordinates required.

Reasoning without geometry: the composition table

If I know $x\ R\ y$ and $y\ S\ z$, what can $x\ ?\ z$ be?

known	known	forced / allowed
x PP y	y PP z	x PP z (*forced*: parts of parts are parts)
x PP y	y DR z	x DR z (*forced*: inside something disjoint)
x DR y	y DR z	anything at all (*free*: 5 options)

- All 25 cells form the composition table — pure algebra, derived once from set semantics, then used forever.
- Abstract semantics: a "model" is any relation assignment closed under this table — *space itself has left the building*.
- Every probe in this project re-derives the table from scratch, so the algebra can never silently drift.

A description logic: you write definitions and facts

A TBox is a set of concept *definitions*, built from properties and quantified spatial relations:

$$\begin{aligned} \text{CityAtRiver} &\equiv \text{City} \sqcap \exists \text{PO}.\,\text{River} && \text{a city } \textit{overlapping} \text{ a river} \\ \text{GermanCity} &\equiv \text{City} \sqcap \forall \text{PP}.(\neg\text{Country} \sqcup \text{Germany}) && \text{a city in } \textit{no country but} \text{ Germany} \end{aligned}$$

Then we assert a few primitive facts — and nothing more:

$$\begin{aligned} \text{Hamburg} &\sqsubseteq \text{City} \sqcap \exists \text{PO}.\,\text{Alster} && \text{(a city, overlapping the Alster)} \\ \text{Alster} &\sqsubseteq \text{River} \sqcap \exists \text{PP}.\text{Germany} \sqcap \forall \text{PO}.\neg\text{Country} \sqcap \forall \text{PP}.(\neg\text{Country} \sqcup \text{Germany}) \end{aligned}$$

Note what we did *not* say: that Hamburg is a GermanCity, or a CityAtRiver.

. . . and the reasoner *infers* the classification

We never *asserted* that Hamburg is a GermanCity or a CityAtRiver — the reasoner derives both.

City at a river (immediate):

$$\text{Hamburg PO Alster} \wedge \text{Alster} \sqsubseteq \text{River} \implies \text{Hamburg} \sqsubseteq \text{City} \sqcap \exists \text{PO.River} = \text{CityAtRiver}.$$

German city — needs the composition table. For any country c with Hamburg PP c:

- Alster PO Hamburg PP c $\Rightarrow$ Alster $\{\text{PO}, \text{PP}\}$ c $\qquad\qquad$ $\text{comp}(\text{PO}, \text{PP}) = \{\text{PO}, \text{PP}\}$
- Alster overlaps no country $(\forall \text{PO}.\neg\text{Country})$ $\Rightarrow$ Alster PP c
- every country Alster lies in is a Germany $(\forall \text{PP}.(\neg\text{Country} \sqcup \text{Germany}))$ $\Rightarrow$ c is one

So Hamburg *is* a German city — and every hidden fact reduces to concept satisfiability: $C \sqsubseteq D$ iff $C \sqcap \neg D$ unsat.

$\mathcal{ALCI}_{\mathrm{RCC5}}$: the spatial relations *become* the roles

The logic (Wessel 2002/2003):

- $\mathcal{ALCI}$ = the basic DL with inverse roles;
- roles = the five RCC5 relations themselves;
- the composition table is enforced globally;
- EQ = identity (*strong* semantics).

So one can say things like

$$\forall \mathrm{PP}.\, \neg \mathrm{Lake} \qquad \exists \mathrm{DR}.\, \mathrm{Reserve}$$

"no part is a lake", "disjoint from some reserve".

Why this is different
Concrete-domain DLs constrain spatial *attributes* of objects. Here the relations are *first-class roles*: one can quantify over them. None of the known concrete-domain decidability results apply.

Why the standard toolbox dies on contact

Every pair of elements carries a relation $\Rightarrow$ a model is a complete labelled graph, never a tree.

no tree models

Blocking/unravelling — the workhorse of DL decidability — has no tree to work on.

no finite models

$\exists \mathrm{PPI}.\top \sqcap \forall \mathrm{PPI}.\exists \mathrm{PPI}.\top$ forces an infinite ascending part-of tower.

nondeterministic table

Most cells allow several outcomes — composition *constrains* but rarely *determines*.

Creating one new element forces relations to *every* existing element — and a dormant $\forall$ anywhere may wake up because of it.

The landscape: open exactly at the top

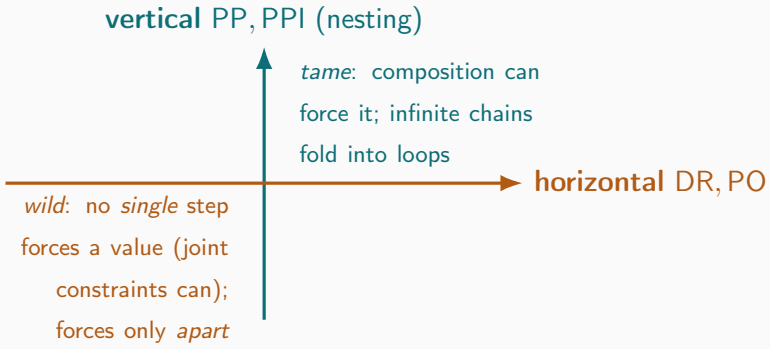
Logic	Relations	Concept satisfiability
$\mathcal{ALCI}_{\mathrm{RCC1}}$	$\{\mathsf{SR}\}$	decidable ($\equiv$ S5)
$\mathcal{ALCI}_{\mathrm{RCC2}}$	$\{\mathsf{DR}, \mathsf{O}\}$	decidable
$\mathcal{ALCI}_{\mathrm{RCC3}}$	$\{\mathsf{DR}, \mathsf{ONE}, \mathsf{EQ}\}$	decidable (via two-variable FO!)
$\mathcal{ALCI}_{\mathrm{RCC5}}$, $\forall$PO-free	RCC5 w/o $\forall$PO	**decidable** (this project)
$\mathcal{ALCI}_{\mathrm{RCC5}}$	all five	**open**; PSpace-hard
$\mathcal{ALCI}_{\mathrm{RCC8}}$	eight RCC8 rel.	**open**; ExpTime-hard

Lutz & Wolter 2006, after proving the *topological* cousins undecidable:

> *"Perhaps the most interesting candidate is* $L_{RCC5}(RS)$ *[*$=\mathcal{ALCI}_{\mathrm{RCC5}}$*] . . . to which the reduction exhibited in Section 8 does not apply."*

Recognized open by the leading experts — for twenty years.

One logic, two completely different axes



- Depth (nesting) is never the problem — ordinary modal-logic folding handles it.
- Everything hard lives in horizontal crowds: many pairwise-distinct neighbours that no finite summary seems to capture.

The asymmetry, made precise (machine-verified)

No *single* step forces a horizontal value. Of the 16 compositions of proper relations, exactly four return a single value:

$$\mathrm{comp}(\mathrm{DR},\mathrm{PPI}) = \mathrm{comp}(\mathrm{PP},\mathrm{DR}) = \{\mathrm{DR}\}, \quad \mathrm{comp}(\mathrm{PP},\mathrm{PP}) = \{\mathrm{PP}\}, \quad \mathrm{comp}(\mathrm{PPI},\mathrm{PPI}) = \{\mathrm{PPI}\}$$

— every one runs through PP/PPI; no single cell yields $\{\mathrm{PO}\}$. But the *intersection* of several can force PO ($\mathrm{comp}(\mathrm{DR},\mathrm{PO}) \cap \mathrm{comp}(\mathrm{PPI},\mathrm{PO}) = \{\mathrm{PO}\}$) — so "shadow" is joint closure, not single-path.

Disjointness is half-deterministic.

downward: $x\,\mathrm{DR}\,y \Rightarrow$ every part of x is DR from every part of y

upward: a superpart of x may be DR, PO, or PPI to y — free

So one high DR edge dispatches an *arbitrarily wide* crowd of disjointnesses below it, for free. What resists: PO, and the upward direction.

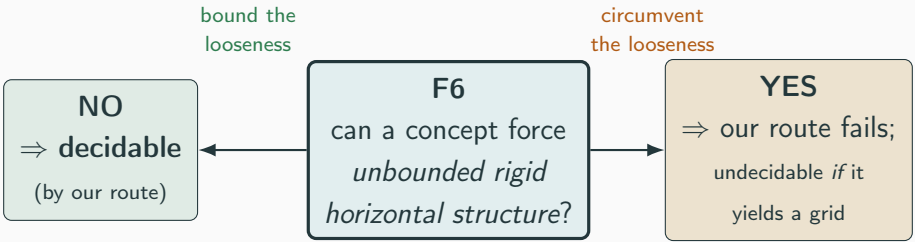
Shadows, live width, and the property our route turns on

- An edge is a shadow if composition already forces it from edges you have — it costs a decision procedure *nothing*.
- The live width of an interface = the number of *non-shadow* edges there: the genuine branching that must be recorded.

Property F6 (the keystone of the certificate route)
Is the live width of models bounded by a computable function of the input concept?

If yes (and two argued steps are proved) ⇒ finite certificate space ⇒ satisfiability is decidable.

The knife's edge: one fact, two futures



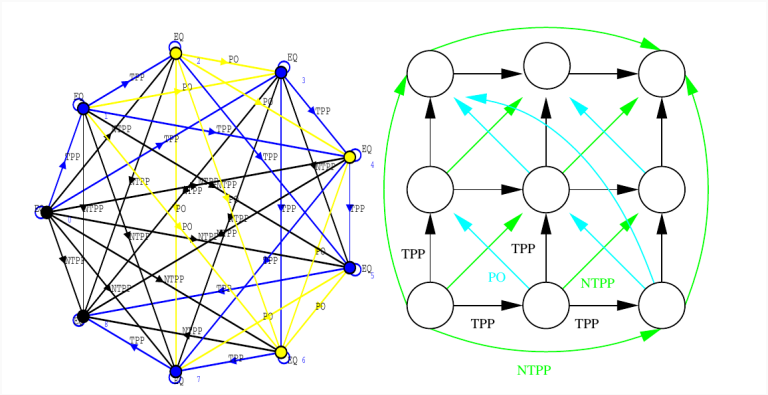
The *same* looseness of the composition table blocks both proofs — a *shared* obstruction, not a theorem that either answer settles the problem. That is why neither direction has moved in twenty years.

The standard weapon needs a rigid grid

- Classic undecidability proofs encode domino tilings of $\mathbb{N} \times \mathbb{N}$: build a grid, force tile constraints, reduce the halting problem.
- The crux is rigidity: going *east then north* must land on *exactly the same cell* as *north then east* ($R_X \circ R_Y = R_Y \circ R_X$, the "coincidence condition").
- Every known route into $\mathcal{ALCI}_{\mathrm{RCC}}$ needs a feature it does not have: number restrictions, arbitrary role axioms, topological rigidity, nominals...

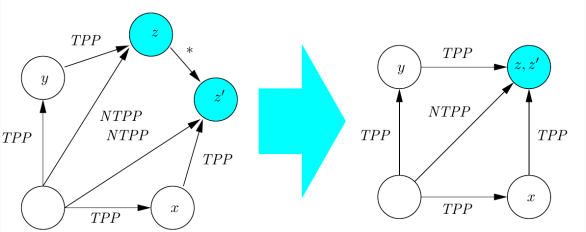
Question (2003 and today): can the composition table itself supply the rigidity?

2003: the grid models *exist*...



An RCC8 network isomorphic to an $n \times n$ grid (report7, Fig. 10, 2002/2003) — the construction *found by a Lisp enumerator* and drawn with CLIM (MCL), not by hand. The models are *out there*.

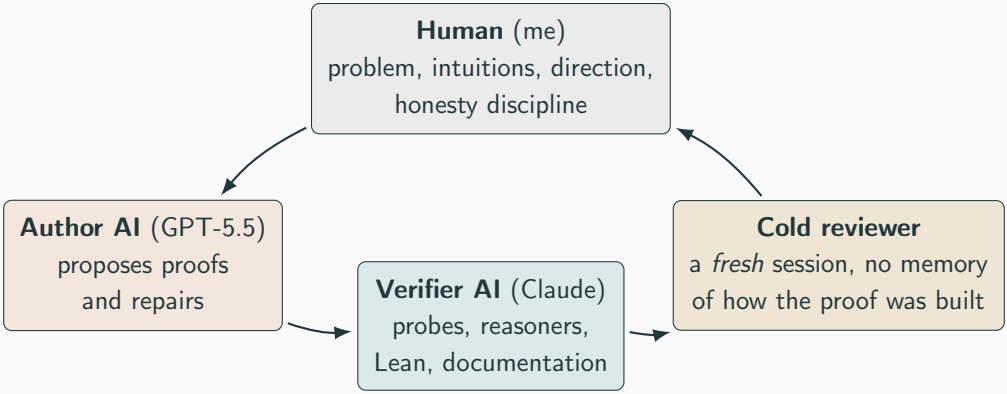
. . . but no known concept *forces* them: the coincidence obstruction



report7, Fig. 11: successors z (east-north) and z' (north-east) *should* be one corner — the table *alone* allows six relations, forcing neither merge nor split. You *can* force the merge (clash out the five non-EQ options, EQ remains) — but that's the trap: those markers collapse *every* corner at the next level (SAT at level 1, UNSAT at 2; RCC5 *and* RCC8). Neither gives a grid: loose $\Rightarrow$ no join, forced $\Rightarrow$ total collapse.

> *"Even though these infinite models do exist, we have found no way to enforce the depicted model(s) by means of a concept term."* (report7, 2003)

The cast, and the loop



Propose → verify → attack → repair. Round after round, March–July 2026.

The campaign at a glance

- ∼4 months, ∼30 repair rounds
- 22 adversarial reviews — a defect or overclaim found in all but *three* (the 22nd found nothing to correct)
- 95 self-contained verification probes (each re-derives the algebra from set semantics)
- 5 Lean files, ∼52,000 lines, zero `sorry`
- 2 working reasoners, cross-validated on a 911-case suite

The most telling pattern
Unsatisfiable concepts *were* accepted — by manuscripts as literally written. Every such case traced to a missing clause in an argument, never to a counterexample to the conjecture itself.

That is why the label is *strongly supported* — and why it is still *not certified*.

Route 1: tableaux with blocking — broken by totality

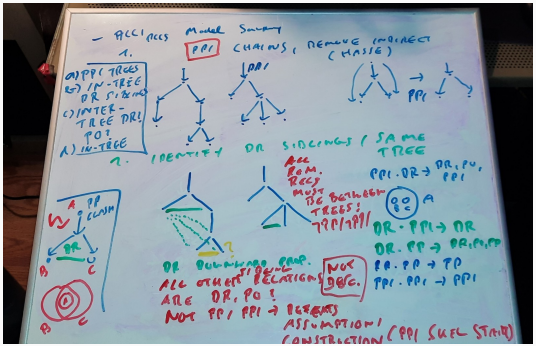
The default DL method: build a model top-down, *block* when a node repeats an ancestor's type.

- Here, composition forces edges into already-built nodes: from $x\,\mathrm{PP}\,y$, $y\,\mathrm{DR}\,z$ the table forces $x\,\mathrm{DR}\,z$.
- Diagnostic: $C_{\mathrm{force}} = \exists \mathrm{PP}.(\exists \mathrm{DR}.A) \sqcap \forall \mathrm{DR}.\neg A$ is **unsatisfiable** — any procedure that ignores forced edges wrongly accepts it.
- Dormant $\forall$'s reawaken blocked nodes; no fixed-arity local summary we tried (types, triangles, quadruples, profiles, ...) was enough.

Lesson: in a total-relation logic, "local" does not exist.

Whether a node may be reused depends on the whole graph — a *global* condition no local test sees.

Route 2: the split-forest idea (a whiteboard, April 2026)



- Present the complete graph as a *forest of part-of trees*: split shared parts into EQ-mates, re-orient vertical edges into the trees.
- All remaining cross-edges are horizontal; complete them by the RCC5 *patchwork property*.
- The one component every cold review found sound — though the certified route keeps only its *shape*, not this machinery: no patchwork appeal, no quotient.

The human contribution: this decomposition. The models made it precise.

Route 3: tree automata — "acceptance without validity"

Package the search as a parity tree automaton (emptiness is decidable)...

- ...but the input tree omits the relations between fresh witnesses and the sources of $\forall$-constraints.
- An *alternating* automaton explores in independent copies — and copies guess the missing relations independently.
- A cold review built the killer: two copies privately guess $\{\mathsf{PP}\}$ and $\{\mathsf{DR}\}$ for the same pair; $\mathrm{comp}(\mathsf{PP},\mathsf{PP}) \cap \mathrm{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{PP}\} \cap \{\mathsf{DR}\} = \emptyset$ — every copy accepts, no model exists.

So the automaton was abandoned, in favour of explicit finite certificates checked by composition tests — with a *truth lemma* instead of an acceptance condition. (This is what got formalized.)

Machine-checking: what the Lean kernel buys

- Lean 4: a proof assistant whose kernel type-checks *every* inference; a proof with a gap *does not compile*.
- Our discipline: Lean 4 core only — no libraries; every object built from first principles; tiny trusted base.
- "zero `sorry`" = no admitted gaps anywhere.

Why we moved to Lean

Most reviews broke a *prose* proof — typically at a step that was *assumed*, not proved. The kernel cannot be talked past. After the move, reviews stopped finding soundness defects and found *interface* gaps and overclaims instead — each closed or withdrawn.

What exactly is certified (zero `sorry`)

Result	Meaning	File
Soundness pipeline	valid certificate ⇒ real RCC5 model	`Round19Transport`
Abstraction faithful	satisfiable ⇔ Hintikka-realizable	`Round19Transport`
RCC5 normal form (both dir.)	network ⇔ order + disjointness	`RCC5NormalForm`
Forcing reduction (cond.)	F6 fails ⇔ forced crowds, *if* adequacy	`ForcingReduction`
Dovetail schema (cond.)	qual. F6 ⇒ decidable, *if* searches complete	`SemiDecidability`
∀PO-**free decided** (2026-08-29)	a *decision procedure*, raw input	`POFreeLift`
PO-erasure ext. (2026-10-08)	decides a slice *with* ∀PO	`POFreeLift`
Abstract = concrete	composition table ⇔ real sets (cf. L&W Thm 23)	`POFreeLift`

Not in the kernel: F6 itself, and the steps from it to a complete certificate search. They appear only as *hypotheses* of conditional theorems — never axiomatized, never proved.

The full-logic theorem (conditional; its premise is open)

Conditional decidability — what the kernel actually checks
If a verified finite checker is complete for some computable list of candidate certificates per concept, **then** $\mathcal{ALCI}_{\mathrm{RCC5}}$ concept satisfiability is decidable.

- The soundness half is kernel-checked end to end: certificate syntax → closure conditions → proper RCC5 frame → truth lemma → *actual model of the input concept*.
- The certificate checker is a real executable Boolean function — no oracle anywhere (a previous version had a hidden one; a cold review caught it; it was rebuilt).
- Open: the premise. Our route to it has two *argued, unproved* steps: F6 $\overset{?}{\Rightarrow}$ refined W2′ $\overset{?}{\Rightarrow}$ complete lists.

Unconditional win #1: a decidable fragment — now *certified*

Theorem (∀PO-free fragment), machine-checked 2026-08-29
Concepts whose NNF has no ∀PO are decidable. In Lean, with fragment membership the *only* hypothesis: `decidableFSat` (raw input) `decidableSat_cone` (NNF) `decidableSetSat` (real sets)

- Still allowed: ∃PO, ∀DR, ∃DR and all part-of modalities — an expressive spatial fragment.
- On raw input the condition is *polarity-sensitive*: no ∀PO positively, no ∃PO *negatively*.
- Argued by two routes, certified by a third — GPT-5.6 Sol's *cone scheme*. Now its own paper.
- Concept satisfiability only (not closed under negation); decidable $\neq$ runnable — doubly exponential.

Why *that* fragment — the one-sentence reason

The residual default
When composition leaves a pair's relation open, you can always settle it as PO — *unless some formula can object to a* PO *edge*. The only formula that can is $\forall \mathrm{PO}.D$.

- So in the fragment the lazy arrangement — declare only what a birth forces, default every remaining pair to PO — is always a model.
- Hence nothing ever has to be coordinated: every relation between two different children is the residual, and the residual is free. That is why the construction needs no joint realizability.
- Put $\forall$PO back and you do not lose one case of one induction — you lose the default itself. That is the whole boundary.
- The arrangement it builds is a split forest — part-of trees, horizontal relations read off: Route 2's whiteboard idea, *built* rather than extracted.

Is the fragment new? What the literature actually says

Lutz & Wolter 2006 — the one place anyone gestured at this
They leave $L_{\mathrm{RCC5}}(RS)$ — *exactly* this logic — open, and float: "giving up some of the relations... dropping PO still results in a useful formalism".

- That sentence defends usefulness, not decidability — no proof, no sketch, no conjecture.
- Our restriction is weaker: PO stays in the language. $\exists$PO is kept; only $\forall$PO goes.
- What they *did* prove about it — r.e., representable by regions (Thms 23, 24, 28) — says nothing about decidability.
- Wessel's 2003 report canvassed *stronger* restrictions, all restoring the finite model property — which this fragment lacks.

New so far as we can determine — deliberately weaker than a priority claim.

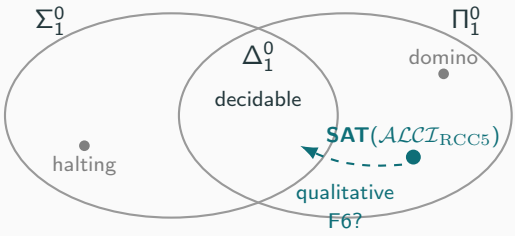
Unconditional win #2: the RCC5 normal form

Theorem (both directions kernel-checked, arbitrary domains)
A strong-EQ RCC5 network is composition-closed *iff* it is an ordered-disjoint structure:

- PP is a strict partial order (a genuine part-of hierarchy);
- DR is a symmetric, downward-closed disjointness;
- comparable pairs never disjoint; PO is the residual.

Every RCC5 model, anywhere, *is* just: a hierarchy + a downward-inherited "keeps-apart" relation + overlap as the leftover. Both directions are kernel-checked for arbitrary domains, the "$\Leftarrow$" with GPT-5.6 Pro's canonical set representation. That RCC5 structures are representable by regions at all is Lutz & Wolter's Thm 23 (2006); what is new is this form, the explicit construction, and the kernel proofs.

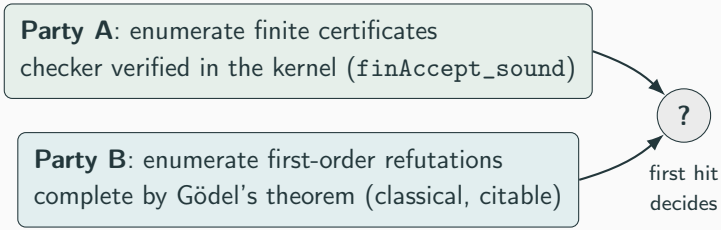
A rediscovery: the problem's address is Π_1^0



- The whole semantics is finitely first-order axiomatizable; by Gödel completeness, *un*satisfiability is enumerable.
- So SAT is co-r.e. (Π_1^0) — the domino problem's *ceiling* (membership; we don't claim hardness).
- Not new: Lutz & Wolter, 2006 (Thm 28). We rederived it and at first claimed it — since corrected.

Consequence: half of the keystone was never needed

A Π_1^0 problem is decidable *iff* SAT is also enumerable. So run two search parties and wait for the first hit:



- For this route, needed: only qualitative F6 — *some* finite certificate per satisfiable concept, no computable bound.
- The bound the reviewers kept demanding (and that kept breaking) comes back for free, afterwards — a theorem of the schema.

"Isn't this just first-order logic?" — yes, and it helps

$\mathcal{ALCI}_{\mathrm{RCC5}}$-SAT *is* FO satisfiability of a finite theory. Does it land in a decidable fragment of FO?

Fragment	Fate	Why
two-variable FO^2	×	composition is ternary (RCC3 *did* fall this way in 2003!)
prefix classes	×	unbounded $\forall\exists$ alternation, FMP
guarded / clique-guarded	×	totality is the one unguardable axiom
unary/guarded negation	×	finite model property; totality again
guarded + transitivity	×	undecidable outside guards
FO^2 + 1 transitive rel.	**?**	decidable; a 2-copy encoding *nearly* fits

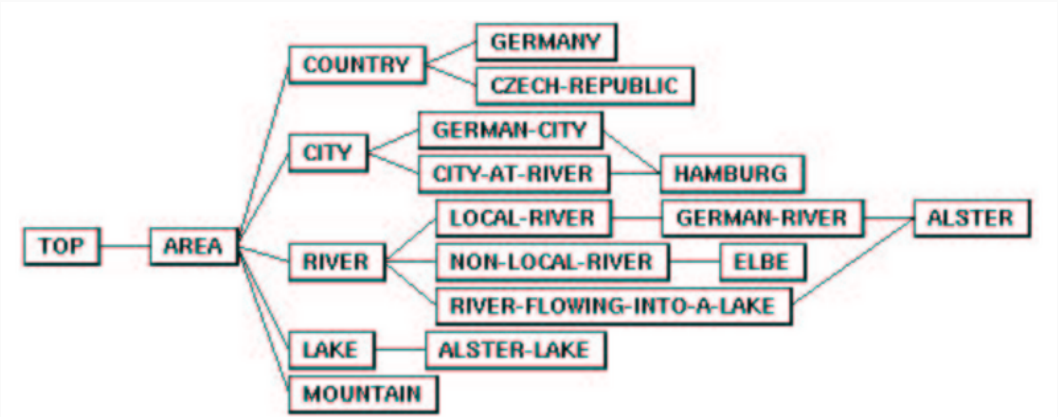
The failure that matters traces to one fact: every pair is related, so models are maximally non-local — the obstruction F6 names.

A working prototype reasoner (validated, not proved)

- The cover-tree tableau ($\sim$350 lines of Python): exact-occurrence blocking on the split-forest foundation.
- A 911-case suite (99 labelled, 812 oracle comparisons) — zero mismatches; 775/775 decomposition tests. One latent gap, found by code review (April), fixed without changing a verdict.
- One wrong verdict (found Nov. 2026): it rejects a satisfiable concept whose two overlap-witnesses must be *nested*. Incomplete, not unsound.
- Decides correctly even the known trap (the strong-EQ PO-loop) that fools type-based procedures.

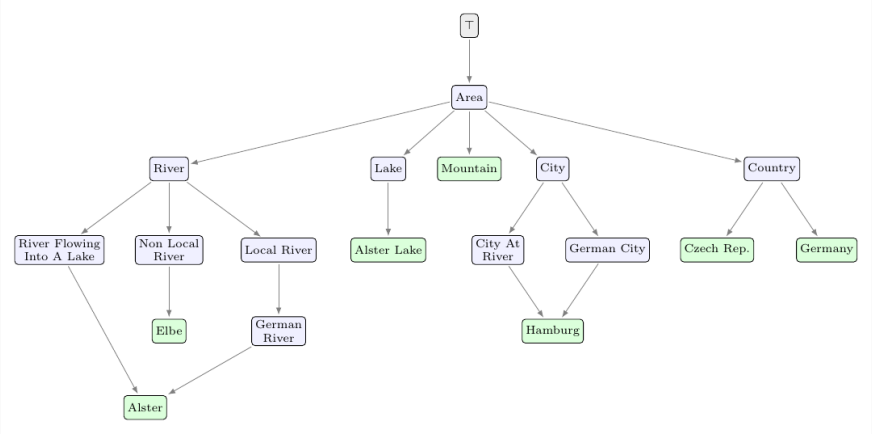
A running algorithm and a kernel-checked surround, pointing the same way — evidence, not a proof.

2003: a prototype reasoner



The GIS concept taxonomy in report7 (Fig. 6), 2003 — itself *computed by a prototype reasoner* of the era, not drawn by hand.

2026, the cover-tree reasoner: all 21/21 — edge for edge



The 2026 cover-tree tableau reproduces it exactly — two reasoners, 23 years apart, identical result. (Green = individuals; *Hamburg* and *Alster* each have two parents — a DAG, not a tree.)

What the models did — and what they could not

Demonstrated

- months-long rigorous formal development, ∼30 rounds
- genuinely useful *adversarial* review (with machine-checked witnesses)
- a ∼52,000-line kernel-checked Lean development
- working, validated software

Not demonstrated

- the decisive novel idea — F6 is uncracked
- reliable *self*-review (both models declared victory prematurely; both were caught)
- working unsupervised: the human directed every round

Neither model is self-sufficient. What carried the project was the loop, not any single model.

The failure mode worth naming: productive-looking thrashing

Three weeks. Four successive repairs to one architecture. Each locally reasonable, each probed, each refuted by an exact finite countermodel — and it proposed the next one.

- It never concluded the class was wrong — though its own design document already recorded this as the *fifth* occurrence of the same shape.
- Not hallucination, not local error: every step was correct and checked. The context makes the framing feel settled because it was never questioned there.
- What broke it came from outside: a halt, and a different model — no history in the design — asked for a different architecture. That answer became the certified route.

Cold review, aimed at *design* rather than proof. Knowing *when* to call it was the human's — we cannot automate that.

The method that worked (steal this)

1. **Cold review beats hot review.** A *fresh* session with no memory of the construction found what warm reviewers missed — every time it mattered. Make it a standard step.
2. **Don't trust — verify.** We cross-checked load-bearing claims mechanically: 95 probes that re-derive the algebra from set semantics, and ultimately the Lean kernel. The machine checks are the ground truth, not the model's confidence.
3. **Label honestly.** *Strongly supported* $\neq$ *certified*. Overclaims kept slipping in — most recently in our own status tables — and cold reviews kept catching them; each was withdrawn on the record.

The full timestamped conversation log is public — every idea, defect, and repair traceable to its origin, human or model.

If nothing else: a message to fellow researchers

- Whatever the fate of the mathematics, this campaign is an existence proof of what frontier models could already do in serious formal research — *in July 2026*.
- Anyone doing serious logic or mathematics should know these capabilities exist. Ignoring them does not advance science; declining to use them is simply unwise.
- Saying exactly this was a stated purpose of the work. Its first outing (DL 2026) was rejected without engaging either the mathematics or the message — so it goes out in the open, with the full audit trail.

The instruments have changed. The standards — verification, adversarial review, honest labels — matter more than ever.

The results, by verification level

Level	Results	What it means
Lean-certified	∀PO-**free decidability** (raw, NNF, sets; + the PO-erasure slice *containing* ∀PO); abstract = concrete; soundness pipeline; faithfulness; normal form; conditional reductions *as conditionals*	kernel-checked, stated axioms
Published	Π^0_1 membership; representation by regions (Lutz & Wolter 2006, Thms 23, 24, 28)	an external theorem, cited
Theorem-level	the fragment's two *arguments* ($K(C_0)$, extraction); W2′ refinement	proof in a work package
Empirical	cover-tree tableau; 911-case suite; model verifier	test evidence, not a decision theorem
Open	F6; whether identity-selector minors are its only obstruction; completeness of the certificate scheme; full decidability	no proof claimed

Where it stands

Full decidability: open

our route's crux: can a concept force unbounded *identity-selector* structure? (conjectured)

- NO ⇒ our route can close; YES ⇒ it fails, and undecidability would still need a grid.
- One sub-problem off the pile: ∀PO dropped — decided in the kernel (own paper); extended 2026-10 to a slice containing ∀PO.
- Reproducible: 5 Lean files (∼52k lines, 0 `sorry`), 95 probes, 2 reasoners, full log.

Open since 2003. Still standing — but one wall has come down.

`github.com/lambdamikel/alcircc5` — thank you!

Backup: the full RCC5 composition table

comp	DR(a,b)	PO(a,b)	EQ(a,b)	PPI(a,b)	PP(a,b)
DR(b,c)	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b,c)	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
EQ(b,c)	DR	PO	EQ	PPI	PP
PP(b,c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b,c)	DR	DR, PO, PPI	PPI	PPI	$*$

Row $= S(b,c)$, column $= R(a,b)$, entry $=$ possibilities for (a,c); $*$ $=$ all five. Re-derived from finite set semantics by every probe; matches the certified Lean table cell for cell.

Backup: what a finite certificate looks like

- **Catalogue** (the finite blueprint):
  - a *core network* — the non-repeating part of the model;
  - finitely many *attachment templates* — recipes for gluing a fresh witness cluster onto an interface;
  - a *steering function* fixing the horizontal relations from old occurrences to fresh ones (the only real freedom).
- **Plan**: which template to apply where; *unfolding* it (possibly forever) produces the split-forest model.
- **Soundness**, kernel-checked: one finite check on the catalogue guarantees *every* unfolding is a legal RCC5 model.
- **Live width** = live occurrences that must coexist at one interface during unfolding — the quantity F6 must bound.

Backup: the sharpened F6 — identity-selector minors

- The naive ways to force width (shell half-graphs, prefix pumps, PO-gaps) all collapse to finite catalogues — probe-checked.
- The sharp obstruction: n pairs (L_i, U_i), each kept apart by a private selector A_i and by nothing else — the selector incidence matrix is the $n \times n$ *identity*.
- A finite catalogue can hold only boundedly many private selectors. **Conjecture:** F6 fails iff some concept forces such minors for every n.
- Bare RCC5 networks with unbounded such structure *exist* (machine-checked) — so any F6 proof must use $\mathcal{ALCI}_{\mathrm{RCC}}$*-forceability*, not RCC5 consistency alone. The uniformization side-condition (W2′) is argued to fold into F6.